

In this exercise we will continue with the VMC simulations on the transverse field Ising model with periodic boundary conditions:

$$\mathcal{H}_{\text{TFI}} = J \sum_i \sigma_i^z \sigma_{i+1}^z - \Gamma \sum_i \sigma_i^x \quad (1)$$

using more sophisticated variational Ansätze (NQS) compared to the last exercise.

As we have seen in that exercise, where we used a "simple" mean-field and a Jastrow Ansatz, it can be very cumbersome to explicitly compute and implement the derivatives of variational Ansätze.

Especially in the case of more complicated functional expressions and a large number of parameters it becomes infeasible to do these computations by hand. As an alternative we could use finite differences to estimate the relevant derivatives. However, as we have seen in the first exercise session, these estimates depend a lot on the function and the discretization that is chosen.

The most reliable technique to compute derivatives automatically is called *Automatic Differentiation* and is implemented in the open-source package [JAX](#).

In all the following exercises we will use the open-source package [NetKet](#) which provides functionality to write your own neural network variational Ansätze and allows the simulation of many-body spin systems using the VMC algorithm (and much more). Additionally since it uses JAX as foundation, automatic differentiation is directly provided for us.

As in the last exercise we will use real variational Ansätze and we work in log-space.

Problem 8.1 MLP Ansatz

In a first step we will use a simple neural network architecture, called multi-layer perceptron (MLP), that is formally written down as:

$$\log(\psi_\theta(s)) = \phi(W^{(L-1)}x^{(L-1)} + b^{(L)}) \quad (2)$$

$$x^{(l)} = \phi(W^{(l-1)}x^{(l-1)} + b^{(l)}) \quad (3)$$

where $x^{(0)} = s$ is the initial configuration, you sample from the squared wave-function, L is the total number of layers in your Ansatz, ϕ is a (nonlinear) activation function and W and b are variational parameters of the Ansatz. To see a graphical representation of the above expression, see [here](#) (with the difference that our neural network only has a single unit in the output layer). In this exercise we will use the *ReLU*-activation defined by $\text{ReLU}(x) = \max(0, x)$.

- Using JAX and FLAX, implement the above variational Ansatz such that it takes a batch of samples of the form $(n_{\text{samples}}, n_{\text{dims}})$ and output the variational Ansatz evaluated for all samples $(n_{\text{samples}},)$.
- Use the template and your variational Ansatz to perform a VMC simulation of the above TFI system for $N = 20$ spins.

- c) Repeat b) with networks of different widths. Start with a width of $N/4 = 5$ hidden neurons and one hidden layer. Rerun the simulation with $N/2 = 10$ hidden neurons etc.. Examine how your result (the energy you obtain) depends on the width of the neural network in the variational Ansatz you use. Also increase the depth of the network, what can you observe?

Problem 8.2 (Optional) Convolutional Neural Network Ansatz (CNN)

A more sophisticated variational Ansatz is given by a convolutional neural network, see [here](#) for an overview.

Repeat the above exercise with a CNN Ansatz. Analyse the quality of your results w.r.t. to the size of the filter you use as well as w.r.t. the pooling function in use. Also compare the energy you obtain with the energy you found in the above exercise. Can you explain the difference?